

Firewall, IDS, and IPS Implementation Report

AI-Agentic SOC Triage & Investigation Engine

1. Introduction

The AI-Agentic SOC Triage & Investigation Engine is designed to automate the initial investigation of cybersecurity alerts by collecting and correlating evidence from multiple security sources. The project uses an alert-driven workflow in which security events are analyzed before being presented to a human analyst. The system is designed to correlate identity, SIEM, and endpoint evidence and produce an auditable investigation report.

As part of the security controls implemented in the lab, a **firewall rule**, **Intrusion Detection System (IDS)**, and **Intrusion Prevention System (IPS)** were configured to provide layered network protection and generate security events for investigation.

2. Firewall Rule Implementation

Firewall: UFW

The Ubuntu host firewall was configured using **Uncomplicated Firewall (UFW)**.

Implemented Rule

Rule: Allow SSH traffic only on TCP port 22.

```
sudo ufw allow 22/tcp
```

```
sudo ufw enable
```

```
sudo ufw status verbose
```

The resulting rule permits inbound SSH connections while allowing the host firewall to control other unsolicited inbound traffic.

Purpose

The firewall provides the first layer of defense by controlling which network connections are permitted to reach the system. In this implementation, SSH access was explicitly allowed because it is required for remote administration of the security lab.

Security Benefit

This follows the principle of limiting network access to required services rather than exposing unnecessary ports. The firewall therefore reduces the host's externally accessible attack surface.

3. IDS Configuration

IDS: Suricata

Suricata was configured as the network Intrusion Detection System.

The IDS monitors network traffic and compares observed activity against detection rules. Unlike the firewall, the IDS is designed to identify suspicious or malicious activity rather than simply allow or deny connections.

A basic Suricata configuration was used to monitor the lab network and generate alerts for suspicious traffic.

Example configuration:

af-packet:

- interface: eth0
- cluster-id: 99
- cluster-type: cluster_flow
- defrag: yes

Suricata rules were enabled to identify suspicious network activity.

Purpose

The IDS provides visibility into network activity that may indicate reconnaissance, scanning, brute-force attempts, or other suspicious behavior.

This supports the project's objective of providing the AI triage engine with security events that can be investigated and correlated.

4. IPS Configuration

IPS: Suricata Inline Prevention

Suricata was additionally configured in **IPS mode** so that identified malicious traffic could be blocked rather than simply reported.

The IPS uses Suricata rules containing the drop action.

Example:

```
drop tcp any any -> $HOME_NET 22 (msg:"Possible SSH Attack Detected"; flags:S; sid:1000001; rev:1;)
```

Purpose

The IPS provides an additional layer of protection by actively preventing traffic identified as malicious.

The distinction between the IDS and IPS configurations is:

Security Control	Function
Firewall	Controls permitted network connections
IDS	Detects and reports suspicious activity
IPS	Detects and blocks suspicious activity

The IPS was configured so that potentially malicious activity could be stopped before reaching the protected system.

5. Example of a Detected Security Event

Event: Suspicious SSH Connection Attempts

A simulated SSH attack was used to demonstrate the detection capabilities of the security controls.

The test generated repeated connection attempts against TCP port 22.

Example Alert

[ALERT] Possible SSH Attack Detected

Source IP: 192.168.1.50
Destination IP: 192.168.1.100
Destination Port: 22
Protocol: TCP
Action: DROP
Detection: Possible SSH Attack

Event Analysis

The event demonstrates how the different security controls work together:

1. **Firewall:** Controls whether the connection is permitted to reach the host.
2. **IDS:** Identifies and records suspicious SSH activity.
3. **IPS:** Uses the detection rule to block the suspicious connection.
4. **SOC Triage Engine:** Can consume the resulting alert and correlate it with additional identity, SIEM, and endpoint information.

The project's architecture is specifically designed around this type of workflow: an alert is received, the investigation engine determines what evidence is needed, evidence is correlated into a timeline, and an analyst receives a risk-focused report.

6. Integration With the AI-Agentic SOC Project

The firewall, IDS, and IPS provide the **security telemetry layer** for the AI-agentic SOC project.

The resulting security alerts can be represented as structured JSON and passed into the investigation engine. The project specifically defines alert ingestion as accepting structured JSON security alerts modeled on SIEM/Sigma-style events. For example:

```
{
  "alert_type": "SSH Attack",
  "source_ip": "192.168.1.50",
  "destination_ip": "192.168.1.100",
  "destination_port": 22,
  "protocol": "TCP",
  "severity": "High",
  "action": "Blocked"
}
```

The AI agent can then determine what additional investigation is required and correlate the alert with historical and endpoint information.

7. Security Control Summary

Control	Implementation	Primary Purpose
Firewall	UFW – TCP/22 allowed	Restrict network access

IDS	Suricata detection rules	Detect suspicious network activity
IPS	Suricata drop rule	Detect and block malicious activity
Detected Event	Suspicious SSH attempts	Demonstrate layered detection/prevention

8. Conclusion

The implementation demonstrates a **defense-in-depth security model** using three complementary controls. The firewall establishes basic network access restrictions, the IDS provides visibility into suspicious activity, and the IPS adds active prevention.

The detected SSH event demonstrates how these controls can generate actionable security telemetry. That telemetry can then feed the AI-Agent SOC Triage & Investigation Engine, which is designed to correlate security evidence, map activity to MITRE ATT&CK techniques, generate a risk summary, and provide an auditable investigation for human review.

Importantly, the project maintains a **human-in-the-loop approval gate**: the AI may recommend containment, but it is not permitted to execute containment actions without explicit analyst approval.

Overall, the implementation demonstrates how firewall, IDS, IPS, and AI-assisted SOC triage can work together as a layered cybersecurity defense.